William Hager

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Education

B.S. in Computer Science, University of Rochester Anticipated May 2028

- GPA: 3.96/4.00
- Dean's Scholarship, All Semesters
- University of Rochester Bond Scholar, 2026
- Coursework: Advanced Algorithmic Design, AI/ML Methods, Formal Computational Models and Limitations, Data Structures and Algorithms, Linear Algebra, Differential Equations

Work Experience

Software Engineer Intern, M&T Bank June 2026 - August 2026

- Developed SDMComms, an internal web portal using **Angular**, **JavaScript** and **ServiceNow** that serves over 90 developers across service delivery management teams, consolidating monthly newsletters, announcements, and platform updates in one place
- Integrated email functionality within the portal that allows administrators to send communications directly to end-users without leaving the ServiceNow platform, reducing managerial communications work by ~50%
- Scoped project from ideation to execution by conducting stakeholder interviews across 5 platform teams to define feature requirements, then translated requirements into a prioritized roadmap and agile sprint backlog executed over 10 weeks

Artificial Intelligence Researcher, Roc-HCI (Human-Computer Interaction) Lab August 2025 - May 2026

- Invented and published the Inverse Turing Benchmark, a **Hugging Face LLM Benchmark** that evaluates model submissions over **557 unique dialogue pairs** on ability to recognize if a given conversation is human/human or human/AI
- First author of InverseTuringBench, a paper outlining the Inverse Turing Benchmark and related findings regarding semantic and statistic AI-text detection amongst 29 detection methods ***(EMNLP/LUHME 2026)***
- Contributed to development of *HAL* (Human Aligning LLM), a framework for **LLM alignment** with emphasis on inducing human-likeness traits, increasing win-rate on a human-likeness Turing Test from 34% to 61%

Projects

Runesworn March 2025 - Present

- Designed and developed a **dice-rolling RPG video game** with Godot engine and GDScript (**Python**) to implement 24 unique turn-based combat encounters, 3 playable characters, and 14 customizable dice
- Used Godot's built-in **scene architecture and signal system** to handle transitions between 7 game scenes, managing tooltips and global state using object-oriented principles
- Optimized **no-SQL database and data retrieval system** by implementing a modular management layer, reducing load time by ~66% and improving player experience
- Established and adjusted core game mechanics with a small-scale playtesting group of around 10 early users in order to ensure balanced and enjoyable gameplay, as well as dynamic difficulty scaling

University of Rochester BAJA SAE Website December 2025 - May 2026

- Architected and implemented a full stack web application that averages ~100 weekly visits for the University of Rochester BAJA Society of Automotive Engineers club, using a **React + React Router + MongoDB** stack
- Created 6 idempotent **RESTful** endpoints using **JavaScript and Netlify serverless functions** to ensure dynamically scaled performance based on current user loads
- Built Auth0 protected **schema-based Content Management System**, allowing non-coding club leaders to update content without code changes

CatGPT May 2026

- Engineered CatGPT, an application that uses an open-source transformer model (**Xenova MiniLM, via Hugging Face**) to extract 768-dimensional vector embeddings from user-submitted text input
- Implemented **cosine similarity comparisons** between generated embeddings and 10 pre-computed embeddings for food-related anchor words to quantify semantic relevance on a continuous numeric scale
- Devised an **exponential scaling function** to convert similarity scores into a variable number of output "meows," translating embedding-based similarity into an interpretable result

Skills

Programming Languages: Java, C, Python, JavaScript, GDScript, HTML, CSS
Frameworks: React, Angular, Gradio, scikit-learn, Java Swing, Java AWT, JavaFX, LibGDX
Technical Skills: MongoDB, Git, Linux/Unix, Valgrind, Object-Oriented Programming, LLM Alignment, ServiceNow